

Surface modification of polypropylene waste by the radiation grafting of styrene and upcycling into a cation-exchange resin

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ABSTRACT: In this study, we radiation-grafted polypropylene waste (PPw) powder obtained from the accelerated aging of pristine polypropylene with styrene (St) using γ rays from a Co-60 source at dose rate of 4.78 kGy/h. The influence of the variation of the γ absorbed dose, solvent types, and monomer concentration on the grafting yield was investigated. Solution grafting in methanol resulted in a higher grafting yield of 162.6% at 30 kGy in contrast to the 69.5% obtained in the absence of solvent. Grafting was found to be enhanced with the increased dilution of St with methanol. The grafted St units were incorporated with sulfonic ($-\text{SO}_3\text{H}$) groups by a sulfonation reaction with sulfuric acid. Fourier transform infrared spectroscopy, X-ray diffraction spectrometry, scanning electron microscopy/energy-dispersive X-ray spectroscopy, and thermogravimetric analysis were used to characterize the grafted and sulfonated products. Sulfonation of 162.6% grafted PPw powder exhibited a maximum ion-exchange capacity (IEC) of 3.56 mequiv/g. The utilization of PPw and its conversion into sulfonated PPw with comparable IECs is a new addition to already existing radiation grafting systems. © 2018 Wiley Periodicals, Inc. *J. Appl. Polym. Sci.* **2019**, 136, 47145.

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INTRODUCTION

Polypropylene (PP) is one of leading thermoplastics; it shares about 20% of the total olefin production of the world.¹ It is a low-cost material extensively used to prepare domestic and industrial products because of its distinct mechanical properties, convenient processing, and minimal consumption. In the Asia-Pacific region, the annual demand of PP products is expected to reach 12 million tons by 2018, whereas in the European market, it has become the second largest polymer from a sales point of view.^{2,3} Most PP products are produced without fillers and undergo degradation by various factors, including heat, light, and oxygen. Such products have a reduced service life and thus end up as plastic waste. Such products can easily be collected and have exhibited good potential for recycling or upcycling. Some researchers have converted them into melt-blown nonwoven fibers⁴ and monomers by pyrolysis,⁵ composite formation,⁶ and upcycling into graphene flakes. An appreciable amount of untreated polypropylene waste (PPw) is either incinerated or dumped in landfills; this poses severe negative threats to the global environment.⁷ According to the U.S. Environmental Protection Agency, approximately 20% of municipal solid waste is composed of products containing PP, but the recycling rate is less

than 1%.⁸ Thus, it is of utmost importance to introduce new, cost-effective, and greener methods to upcycle this waste into value-added products.

Radiation grafting has emerged as an effective method to modify polymers; it introduces specific functional groups onto polymer materials by grafting onto specifically selected monomers for desired applications. It is an ecofriendly technology, which requires no chemical initiators, minimizes additives, and uses catalysts with zero activation energy for the initiation of free-radical processes. Various kinds of polymeric adsorbents and ion-exchange resins for the separation and preconcentration of heavy metals from industrial wastewaters have been produced with radiation grafting.^{7,9} Previous studies have demonstrated the radiation grafting of styrene (St) monomer onto different polymers, including PP, polyethylene, and Teflon, to prepare precursor polymers; these were subsequently activated by sulfonation into strongly acidic cation-exchange resins having $-\text{SO}_3\text{H}$ groups.^{10–12} Most commercial ion-exchange polymers are based on St and are available in the form of particles and membranes. However, radiation grafting has enabled the preparation of ion exchangers in other physical forms, such as gels, fibers, and fabrics for special separation purposes. Mostly, pristine polymers

have been used as substrates for radiation grafting and the preparation of ion-exchange materials.¹³ However, because of strict discharge regulations imposed on industries, the preparation of low-cost alternatives has gained impetus. In this regard, with PPw for the preparation of a cation-exchange resin seems interesting. Recently, PP films prepared from PPw were used as a substrate for the radiation grafting of St to be used as an electrolyte membrane for fuel-cell applications.¹⁴ In one of our previous works, PPw powder was successfully converted into an amidoxime adsorbent by radiation grafting; it exhibited good potential for the adsorption of copper ions from aqueous solutions.¹⁵

The main objective of this study was to explore the feasibility to upcycling PPw with a radiation-grafting technique. The simultaneous radiation grafting of St onto PPw was carried out, and the effects of the absorbed dose (kGy), solvent, and monomer concentration on the grafting yield were investigated. The St-grafted PPw was converted into a cation-exchange product via treatment with sulfuric acid (H₂SO₄) under controlled conditions.

EXPERIMENTAL

Materials and Chemicals Used

Isotactic PP (Topilene, J842, Hyosung, South Korea) was used as the pristine polypropylene (PPP) to be model-aged into PPw. St monomer ($\geq 99.9\%$, Sigma-Aldrich) was used without further purification. The other chemicals used, including methanol, isopropyl alcohol, dichloromethane, tetrahydrofuran (THF), hydrochloric acid (HCl), H₂SO₄, sodium chloride (NaCl), and sodium hydroxide (NaOH), were analytical grade. Nitrogen gas (99.999% purity) was used to provide an inert atmosphere. Type I water was used for the preparation of solutions for the determination of the ion-exchange capacity (IEC) of the resin.

Generation of PPw

PPw was obtained by the irradiation of injection-molded samples of PPP by γ rays at 100 kGy and their accelerated thermal aging at 60 °C for 200 h to mimic postconsumed PPP products.^{16–18} The aged specimen pulverized to yield fine powder of 120 mesh was washed with acetone and then vacuum-dried at 60 °C before it was used as a substrate for grafting.

Grafting and Sulfonation

Radiation-grafting experiments were carried out with the use of γ rays from a Co-60 (24 kCi) irradiator having a dose rate of 4.78 kGy/h at room temperature. Precisely weighed fractions of PPw (2.0 ± 0.01 g) were added to separate Pyrex glass vials, each containing 10.0 mL of St monomer or its diluted solution previously deaerated by nitrogen gas. EtOH, methanol, and isopropyl alcohol were used separately as diluents. The contents of the vials were bubbled by nitrogen for a further 10 min and then sealed before irradiation. After grafting at specified absorbed doses, the PPw samples were Soxhlet-extracted with THF for 16 h to extract the homopolymer formed during grafting. The resulting grafted samples (PPw-g-St) were then washed with dichloromethane and dried *in vacuo* at 60 °C until a constant weight was achieved. The grafting percentage was determined with the following equation:

$$\text{Grafting (\%)} = \frac{(w_g - w_o)}{w_o} \times 100 \quad (1)$$

where w_o is the weight of the ungrafted PPw and w_g is the weight of the completely dried PPw-g-St.

The sulfonation of the PPw-g-St was carried out by a previously reported method.¹⁹ Briefly, PPw-g-St particles were immersed in 98% H₂SO₄ in a weight-to-volume ratio of 1:5 in a glass reactor equipped with a stirrer and placed in a water circulation bath. The particles were continuously stirred at 98 °C for 2 h. The mixture was filtered, and the sulfonated residue was washed excessively with type I water until the filtrate was neutralized. It was then washed with 0.2 M HCl; this was followed by repeated washings with water to remove the excess acid. The sulfonated sample exhibited a dark brown color, was vacuum-dried at 60 °C for 26 h until a constant weight was achieved, and was termed as PPw-g-SSt.

Determination of IEC

An acid–base titrimetric method^{20,21} was adopted to determine the IEC of PPw-g-SSt expressed in milliequivalents per gram. A known dried weight of PPw-g-SSt was equilibrated with 25.0 mL of a 3 M NaCl solution for 24 h. The amount of H⁺ ions released from the polymer was determined by the titration of a known aliquot of equilibrated NaCl solution against 0.05 M NaOH with phenolphthalein as an indicator. The IEC was then computed as follows:

$$\text{IEC}_{\text{exp}} (\text{mequiv/g}) = 0.05 \times V_{\text{NaOH}}/w_d \times V_{\text{NaCl}}/V_a \quad (2)$$

where V_{NaOH} is the volume of NaOH solution consumed (mL), w_d is the dry weight of the PPw-g-SSt (g), V_{NaCl} is the volume of NaCl used for equilibration, and V_a is the volume of the aliquot taken for titration.

With the assumption that only one $-\text{SO}_3\text{H}$ group was bonded to the phenyl ring, eq. (3) was used to calculate the theoretical ion-exchange capacity (IEC_{th}) of PPw-g-SSt:²²

$$\text{IEC}_{\text{th}} (\text{mequiv/g}) = \frac{9.62G}{100 + 1.77G} \quad (3)$$

where G is the grafting percentage of PPw-g-St. The degree of sulfonation was calculated from ratio of the experimental ion-exchange capacity (IEC_{exp}) to IEC_{th} as follows:

$$\text{Degree of sulfonation (\%)} = 100(\text{IEC}_{\text{exp}}/\text{IEC}_{\text{th}}) \quad (4)$$

Characterization Study

Fourier Transform Infrared (FTIR) Analysis. A Thermo-Nicolet 6700 FTIR–attenuated total reflectance spectrophotometer (Thermo Electron Corp.) was used to identify the functional groups in PPP, PPw, PPw-g-St (with different levels of grafting), and PPw-g-SSt. The vacuum-dried samples were scanned at a resolution of 6 cm^{−1} from 4000 to 500 cm^{−1}. An average of 116 scans/sample were recorded to maintain an optimum signal-to-noise ratio. The FTIR spectrum of PPw was also used to calculate the oxidation index^{23,24} of PPw with eq. (5):

$$\text{Oxidation index} = A_c/A_e \quad (5)$$

where A_c is the area of the peak in the FTIR spectrum of PPw from 1850 to 1650 cm^{-1} and A_e is the area of the peak from 1420 to 1315 cm^{-1} .^{18,25}

X-Ray Diffraction (XRD) Analysis. The structural changes taking place during radiation-initiated grafting were monitored by XRD analysis with a Thermo ARL X'TRA 48 X-ray diffractometer with Cu K α radiation ($\lambda = 0.15406$ nm) operating at 45 kV and 40 mA. Radial scans were carried out in reflection scanning mode at 2θ values of 10–60° at a scanning rate of 1°/min. The degree of crystallinity (X_c) was calculated with eq. (6) as reported elsewhere:²⁶

$$X_c (\%) = \left[\frac{A_{cr}}{A_{cr} + A_{am}} \right] \times 100 \quad (6)$$

where A_{cr} and A_{am} are the areas of the crystalline peak and the amorphous halo, respectively.

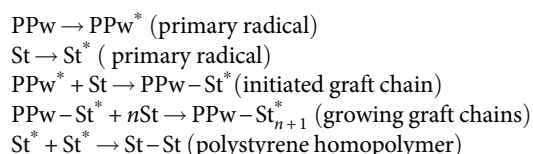
Scanning Electron Microscopy (SEM). The surfaces of the PPP, PPw, PPw-g-St, and PPw-g-SSt particles were scanned with a TESCAN MIRA-3 field emission scanning electron microscope to visualize the morphological variation occurring in PPw that was due to the accelerated aging, radiation-grafting, and sulfonation processes. The samples were sputter-coated with amorphous carbon before they were scanned under an electron beam. An energy-dispersive X-ray spectroscopy (EDX) spectrum of each sample surface was also recorded to reflect the respective elemental composition.

Thermogravimetric Analysis (TGA). A Mettler-Toledo TGA/DSC1 thermogravimetric analyzer was used to carry out the TGAs of PPP, PPw, PPw-g-St, and PPw-g-SSt. The analysis was carried out with 5.0 mg of the dried sample heated from ambient temperature to 800 °C at heating rate of 10 °C/min under a continuous nitrogen gas flow.

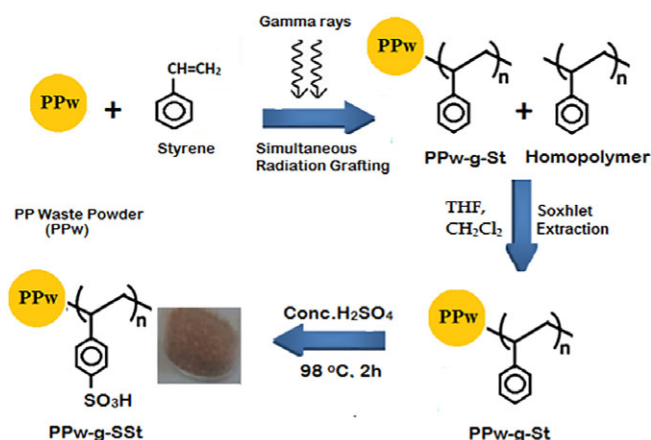
RESULTS AND DISCUSSION

Preparation of the St-Grafted PPw and Its Sulfonation

The photooxidative and thermooxidative treatments of PPP during manufacturing and its service life greatly affects its physico-chemical properties and also lowers its mechanical properties. Generally, the oxidized surface of PPw contains a variety of functional groups, such as ketones, aldehydes, esters, γ -lactones, peresters, and carboxylic acids,²⁷ which may provide active centers for radiation grafting.^{15,28} Because of the radiation sensitivity of PPw, a simultaneous radiation-grafting technique was used. Possible reactions occurring during this grafting process are given next:



where PPw^* and St^* are the primary free radicals of the PPw and St monomer, respectively, which were initiated by γ rays. These radicals further interacted with PPw and St units to initiate



Scheme 1. Steps involved during the radiation grafting of St onto PPw and sulfonation. [Color figure can be viewed at wileyonlinelibrary.com]

grafted chain radicals (PPw-St^*) by grafting-to and grafting-from mechanisms,²⁹ and they continued to grow by free-radical reaction, finally yielding PPw-g-St. In addition, the self-recombination of primary radicals also yielded some homopolymer. Although the higher G (radical) value of PP as compared to that of the G (radical) value of St favored the grafting reaction over homopolymer formation,³⁰ still, the grafted product was treated with THF overnight and washed with dichloromethane to remove polystyrene and PP homopolymer. The resulting PPw-g-St particles were chemically modified with H_2SO_4 via the aromatic electrophilic substitution of the aromatic hydrogen of the grafted St with the $-\text{SO}_3\text{H}$ group. Scheme 1 portrays various steps of grafting and sulfonation on PPw.

Characterization of the Grafted and Sulfonated PPw

FTIR Spectroscopy-Attenuated Total Reflectance Analysis. The chemical changes in PPw with grafting and subsequent sulfonation were studied via FTIR spectroscopy of PPP, PPw, PPw-g-St, and PPw-g-SSt, as shown in Figure 1. Each spectrum contained

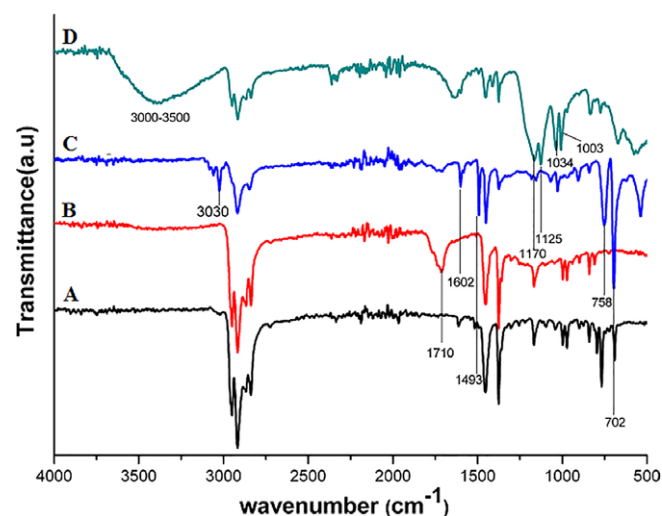


Figure 1. Comparison of the FTIR spectra of (A) PPP, (B) PPw, (C) PPw-g-St (grafting = 162.6%), and (D) PPw-g-SSt. [Color figure can be viewed at wileyonlinelibrary.com]

characteristic bands of PP at 2950, 2862, 2910, and 2834 cm^{-1} due to the alternate symmetric and asymmetric stretching vibrations of aliphatic $-\text{CH}_3-$ and $-\text{CH}_2-$ groups, respectively. The corresponding bending vibrations of these groups were found at 1374 and 1456 cm^{-1} , respectively. In PPw, the appearance of an additional peak at 1710 cm^{-1} indicated the presence of the $>\text{C}=\text{O}$ group from the aging of PPP. The oxidation index of PPw was calculated to be 1.87. After grafting, the spectrum of PPw-g-St displayed some new peaks in the region from 3080–3010 and 1601–1500 cm^{-1} , as represented by both the sp^2 stretching of the C–H and C=C bonds of the grafted St ring, respectively.³¹ The peaks at 758 and 702 cm^{-1} manifested the out-of-plane bending vibrations of the C–H bonds after grafting. However, there was a decrease in the intensity of the peaks at 1710 cm^{-1} ; this indicated the utilization of the oxidized species of PPw for grafting. After sulfonation, the spectrum of PPw-g-SSt showed a characteristic broad band from 3500–3000 cm^{-1} that corresponded to the $-\text{OH}$ stretching vibrations in the $-\text{SO}_3\text{H}$ group. The appearance of a prominent set of peaks from 1170 to 1125 cm^{-1} marked the presence of the C–S bond, whereas the peaks at 1033 and 1003 cm^{-1} represented the stretching and bending frequencies, respectively, of the $\text{O}=\text{S}=\text{O}$ bond of the sulfonic acid group.³²

XRD Study. Figure 2 displays the X-ray diffractograms of PPP, PPw, PPw-g-St, and PPw-g-SSt particles. The presence of characteristic sharp peaks at 14.1, 16.9, 19.0, and 22.1° in the diffractogram of PPP (A) reflected the (110), (040), (130), and (041) planes and depicted its α -monoclinic structure.³³ The X_c of PPP was found to be 42.10%. The diffractogram of PPw (B) revealed no new crystalline peaks, and the peak positions of PPP remained intact. The intensity and shape of the aforementioned crystalline zone peaks indicated that the X_c of PPw was not significantly affected and was found to be 39.91%. The mobility of radiation-induced free radicals was affected because of the slight change in X_c .³⁴ However, after radiation grafting, the reduction in the

height of the crystalline peaks in diffractogram C took place despite the fact there was no change in Bragg's angle. This illustrated the dilution of the crystalline phase of the PPw matrix via grafted polystyrene; this further enhanced the amorphous zone. Thus, the overall X_c was lowered to 21.51%. After sulfonation, the analyzed diffractogram of PPw-g-SSt made it appear to be slightly more crystalline, with an X_c of 23.41%; this could be ascribed to the specific orientation of grafted chains due to hydrogen bonding among closely spaced sulfonic acid groups.

SEM–EDX Analysis. The surface modification of the base polymer due to monomer grafting and sulfonation was examined by SEM. The scanned surface micrographs of PPP, PPw, PPw-g-St, and PPw-g-SSt exposed at a scale of 10 μm and their corresponding EDX spectra are shown in Figure 3. A comparison of Figure 3(A) with Figure 3(C) reveals the smooth and compact surface of the PPP particles, which developed pores, globules, and blisters of various sizes after aging. The creation of porosity and the rough surface was caused by the reaction of radiation and thermally initiated free radicals in PPP with oxygen.³⁴ The increased oxygen content in PPw also confirmed the occurrence of oxidation in PPw. These pores and voids acted as additional grafting sites for the monomer and facilitated the diffusion of the monomer. The reduction in the size of the pores and voids after grafting in Figure 3(E) depicted the hierarchical growth of polystyrene grafts on the active sites created within the pores and on the available smooth surface. The EDX spectrum in Figure 3(F), also revealing an increased carbon content as compared to that of PPw, also confirmed grafting.

Figure 3(G) reveals the patch pattern on the surface of PPw-g-SSt with various smooth patches; this indicated a decrease in the roughness of PPw-g-St due to sulfonation. The decrease in the roughness might have resulted from the hydrogen bonding among the adjacently attached sulfonic acid groups of the St-grafted chains. The appearance of the characteristic K_α signal of sulfur at 2.307 KeV in the EDX spectrum in Figure 3(H) of PPw-g-SSt confirmed the aromatic substitution of sulfonic acid groups on grafted St.

TGA. Figure 4 shows the TGA curves of PPP, PPw-g-AN, PPw-g-St, and PPw-g-SSt. Thermogram A reflects the single-step degradation of PPP starting from 315 °C and completing around 480 °C with the T_{onset} (onset of degradation temperature) residing around 410 °C. The thermal stability of PPw was greatly affected by the thermooxidative treatment, as revealed by the degradation region between 200 and 460 °C in thermogram B. T_{onset} of PPw was lowered by 62 °C because of the earlier volatilization of many end products, which resulted in PPw. After grafting with St, the TGA curve of PPw-g-St manifested increased thermal stability over that of PPw with increasing T_{onset} , that is, to 380 °C. However, the thermogram of PPw-g-SSt showed three prominent mass loss stages. The first decrement in mass occurred in the temperature range 50–180 °C and depicted the loss of water molecules associated with $-\text{SO}_3\text{H}$ groups, which possessed an intense hydrophilic nature.³⁵ The second-stage degradation in the region from 310 to 460 °C indicated the degradation of grafted polystyrene chains.²⁸ The final degradation step

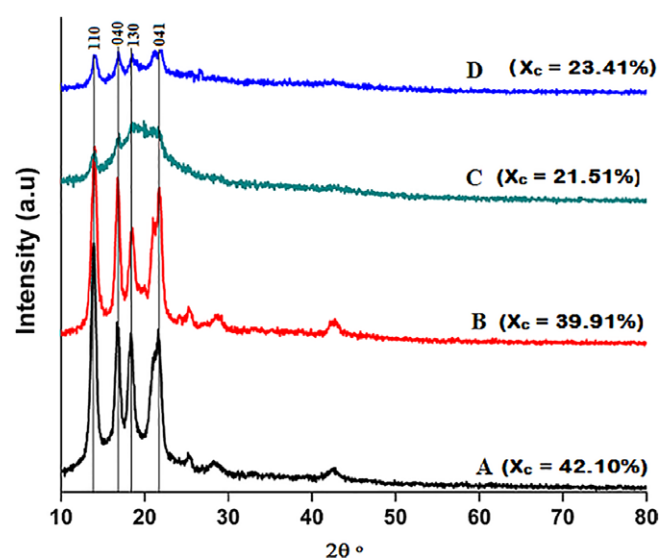


Figure 2. X-ray diffractograms of (A) PPP, (B) PPw, (C) PPw-g-St (grafting = 162.6%), and (D) PPw-g-SSt. [Color figure can be viewed at wileyonlinelibrary.com]

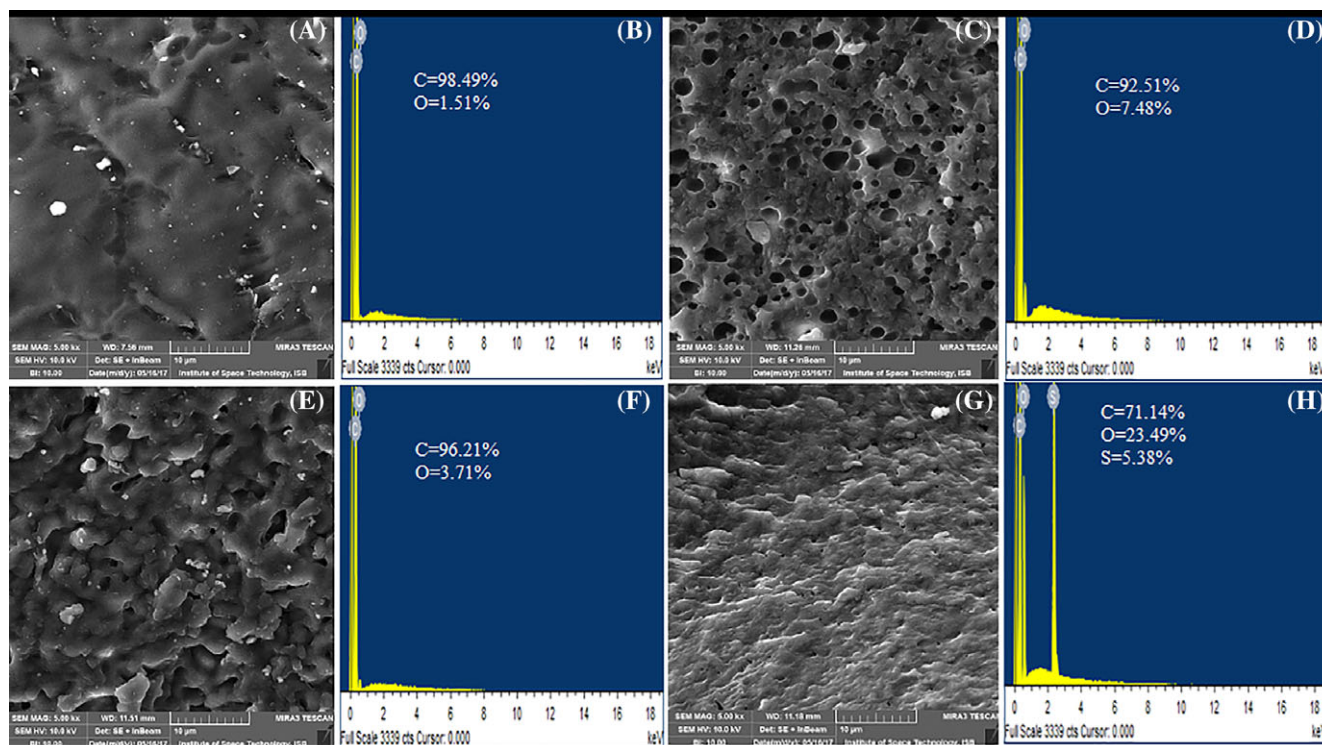


Figure 3. SEM micrographs of (A) PPP, (C) PPw, (E) PPw-g-St (grafting = 162.6%), and (G) PPw-g-SSt. EDX spectra of (B) PPP, (D) PPw, (F) PPw-g-St, and (H) PPw-g-SSt. [Color figure can be viewed at wileyonlinelibrary.com]

corresponded to the degradation of the backbone polymer. The increased quantity of residue may have been due to the sulfur in the residue that could not be volatilized completely until a temperature of 800 °C was reached.

Factors Affecting Grafting

Type of Solvents. In radiation grafting, the degree of accessibility of the monomer to the grafting sites is greatly influenced by the type of solvents used to facilitate monomer transport to the backbone polymer.³⁶ Table I compares the grafting of PPw-g-St

developed by bulk grafting to those developed in the presence of methanol, EtOH, and isopropyl alcohol at 30 kGy. In case of polar protic solvents, such as alcohols, the behavior could be explained in terms of increased radical yield due to the radiolysis of solvent molecules augmented by the increased diffusibility of the monomer to the active sites of the polymer matrix.

The radiolysis of the solvent led to the production of solvated electrons, which increased the radical density in the grafting media, and increased grafting was achieved. The higher grafting in the presence of methanol could also have also been due to the higher dielectric constant of methanol (32.7) compared to those of EtOH (24.5) and isopropyl alcohol (20.2). It has been established that the radiolysis of a monomer with a high dielectric constant favors the production of solvated electrons.³⁷

As a result, the rate of generation of free radicals increased in methanol; thereby, enhanced grafting was achieved. A comparison of the corresponding FTIR spectra of PPw-g-St in bulk to those grafted in solvents is given in Figure 5. The increase in the intensity of the peaks in the marked regions of the spectra also provide evidence for an increased number of St units grafted on PPw in the presence of solvents such as methanol as compared to those grafted in bulk.

Monomer Concentration. Table II shows the relationship between grafting and the St concentration in methanol. It was noticeable that the grafting decreased with an increase in its content in the grafting mixture. Maximum grafting was achieved at a 20% monomer concentration. The grafting not only depended on

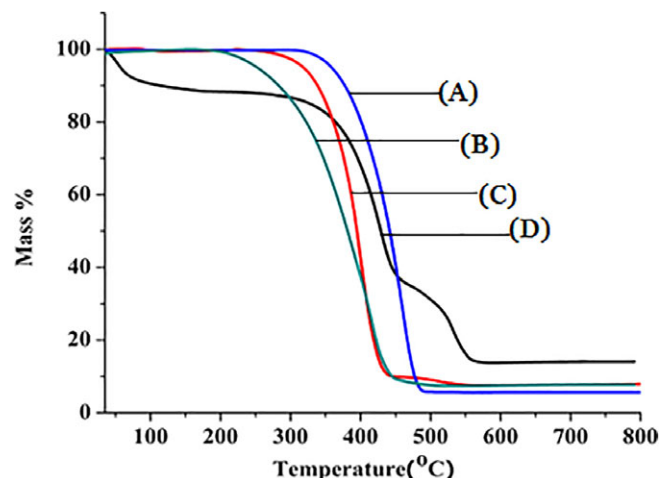


Figure 4. Thermograms of (A) PPP, (B) PPw, (C) PPw-g-St (grafting = 162.6%), and (D) PPw-g-SSt. [Color figure can be viewed at wileyonlinelibrary.com]

Table I. Effects of the Solvents on the Grafting of PPw-g-St at an Absorbed Dose of 30 kGy

Solvent	St concentration (%)	Grafting (%)
No solvent	100	69.5
Isopropyl alcohol	50	59.8
EtOH	50	152.3
Methanol	50	162.6

the amount of trapped radicals at a specific absorbed dose but also on the diffusibility of St through the polymer matrix.

These results suggest that the grafting started at the surface of the particles and proceeded internally by the successive diffusion of St until equilibrium swelling was reached at a 20% monomer concentration.

A comparison of the characteristic identification peaks of St marked in the FTIR spectra of PPw-g-St in Figure 6 at different monomer concentrations also supported the gravimetric results of Table II. With increased dilution of the monomer, the intensity of the characteristic peaks of St functionality in the marked regions of the FTIR spectra also increased.

Absorbed Dose. Figure 7(A) shows the gravimetric estimation of the grafting percentage in PPw-g-St at various absorbed doses in bulk and in a 50% monomer solution in methanol. In mutual radiation grafting, the total number of free radicals created in the mixture were greatly dependent on the absorbed dose. During grafting in bulk, grafting gradually increased with radiation dose, reaching 88.7% at 50 kGy, whereas it increased sharply in the presence of methanol, reaching a maximum of 162.6% at only 30 kGy. The initial rise may have been due to the increase in the number of free radicals generated in the backbone polymer and in the monomer with increasing absorbed

Table II. Effects of the St Concentration in Methanol on the Grafting of PPw-g-St at an Absorbed Dose of 30 kGy

Solvent	St concentration (%)	Grafting (%)
Methanol	20	198.1
Methanol	40	189.5
Methanol	60	140.3
Methanol	80	126.2
No solvent (bulk)	100	69.5

dose. The grafting phenomenon was also accompanied by homopolymer formation.³⁸

Any increase in the absorbed dose beyond 30 kGy increased the homopolymer content; this not only resulted in the depletion of the monomer radicals available for grafting but also resulted in the restricted movement of these radicals to the grafting sites on the backbone polymer; this, thereby, decreased grafting. The corresponding FTIR spectra of the grafted specimens also provided supporting evidence, as discussed earlier.

IEC of PPw-g-SSt

The ion-exchange mechanism involved the replacement of the H^+ ions of $-SO_3H$ moieties with some cations, such as Na^{+1} . Table III indicates the dependence of the IEC_{exp} and IEC_{th} values of PPw-g-SSt on the grafting yield in the cases of both solution and bulk grafting. The increased amount of St units available for sulfonation in highly grafted samples enhances the IEC.¹⁴ Because solution grafting yielded PPw-g-St with a higher degree of grafting, the IEC of the resulting sulfonated PPw-g-SSt samples was found to be greater than that of the sulfonated products yielded from the bulk grafting method.

We also noticed that the results of IEC_{th} and IEC_{exp} at lower degrees of grafting were close; however, with an increase in grafting, the deviation of IEC_{exp} from IEC_{th} increased and caused corresponding decrease in the degree of sulfonation (IEC_{exp}/IEC_{th})

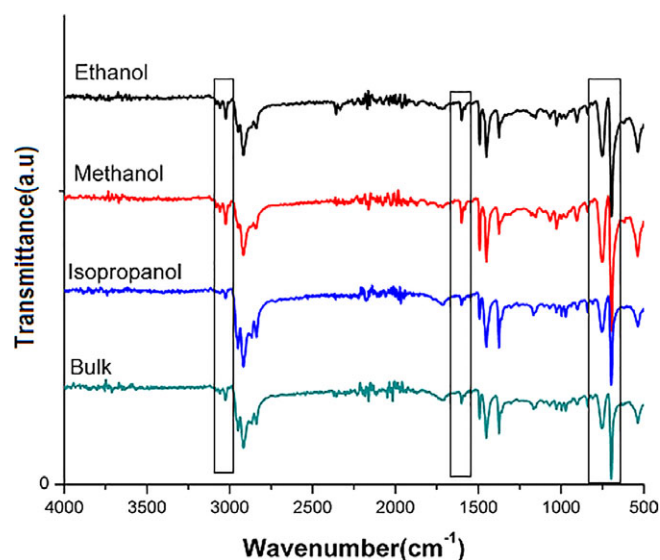


Figure 5. FTIR study of PPw-g-St prepared in bulk and in the presence of solvents (absorbed dose = 30 kGy, St concentration in solvents = 50%). [Color figure can be viewed at wileyonlinelibrary.com]

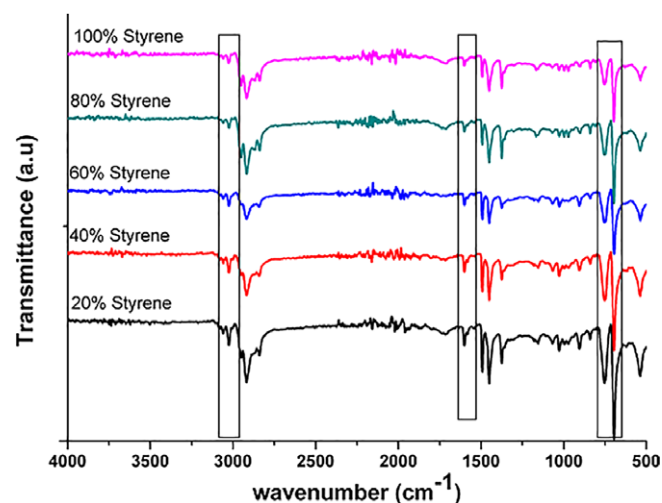


Figure 6. FTIR study of PPw-g-St at different concentrations of St in methanol (absorbed dose = 30 kGy). [Color figure can be viewed at wileyonlinelibrary.com]

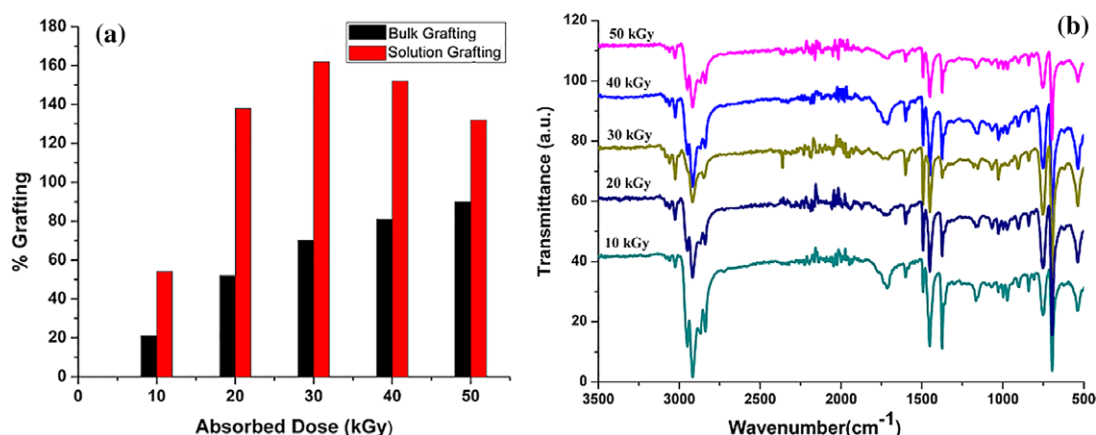


Figure 7. (A) Effect of the absorbed dose on the radiation grafting of St onto PPw at 100% monomer concentration and at 50% dilution with methanol. (B) Comparison of the FTIR spectra of PPw-g-St with 50% methanol-diluted St at various absorbed doses (10–50 kGy). [Color figure can be viewed at wileyonlinelibrary.com]

Table III. Comparison of the IECs and Degrees of Sulfonation in PPw-g-SSt with Various Graftings by the Bulk and Solution Methods

	Grafting (%)	IEC _{th} (mequiv/g)	IEC _{exp} (mequiv/g)	Degree of sulfonation (%) ^a
Bulk grafting	21.2	1.48	1.36	91.5
	44.9	2.40	2.16	89.6
	69.5	2.99	2.69	89.8
	81.5	3.20	2.86	89.1
	89.7	3.33	2.95	88.6
Solution grafting	54.7	2.67	2.46	92.0
	132.1	3.80	3.48	91.3
	138.3	3.86	3.50	90.2
	152.3	3.96	3.54	89.3
	162.6	4.03	3.56	88.2

^a IEC_{exp}/IEC_{th}.

from 92.0 to 88.2% in case of solution grafting, whereas it decreased less in the case of products of bulk grafting.

A comparison of the IEC values of PPw-g-SSt with other sulfonated cation-exchange resins prepared by the radiation grafting of St on other pristine polymers, such as UHMWPE, LDPE, and PP,^{14,39,40} suggests the utilization of PPw-g-SSt resin as a cheaper resin for cost-effective ion separation and water-purification tasks.

CONCLUSIONS

The results of this study indicate that PPw was successfully radiation-grafted with St monomer both in the presence and absence of solvents with γ radiation. The grafting yield was found to increase with absorbed dose in the solution grafting method; this yielded a higher grafting than in bulk grafting. A grafting of 162.6% was achieved by solution grafting in methanol as a solvent at 30 kGy, whereas only 89.7% grafting was achieved up to 50.0 kGy in the absence of solvents. Grafting in methanol was superior to grafting in isopropyl alcohol and EtOH. The sulfonation of PPw-g-St resulted in cation-exchange resins having IECs ranging from 2.46 to 3.56 mequiv/g in solution grafting and 1.36

to 2.95 mequiv/g in bulk grafting. Thus, the prepared PPw-g-SSt could be used for cation-exchange applications for the treatment of water and wastewater.

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